CPSC 483: Introduction to Machine Learning

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Fall 2026

Cal State Fullerton

Review

Midterm Exam

- Rules
  - Electronics → **TURNED OFF** (not on silent/vibrate/sleep/hibernate)
  - Pencil/eraser → write legibly
  - One double-side cheat sheet → Either print or hand-written
- 75 minutes
  - Please arrive ON TIME
- Study the concepts
  - Review the exercise questions
  - Review the slides and read chapters

NO CHEATING!!

Midterm Exam

Will be covering everything until last week

<u>Three parts/types of questions</u>

- Multiple choice
- Short answer questions
  - i.e., True/False (*Provide Reasoning!!!*)
  - Conceptual
- Long answer questions
  - i.e., Scenarios-based

Chapter 1 - Machine Learning Landscape

- What is Machine Learning?
- Why Use Machine Learning?
- Examples of Applications
- Main Challenges of Machine Learning
- Taxonomy of machine learning
  - Distinguishing between Classification/Regression/Clustering
  - Self-Supervised vs. Semi-Supervised Learning
  - Reinforcement Learning

Chapter 2 - Data Preprocessing

- Exploratory Data Analysis
- Test-train split
  - Training/test dataset
  - Instance vs. Label/Class
  - Features/Feature vector
- **Feature transforms**
  - Numerical vs Categorical variables
  - Converting numerical variables to categorical
    - Equal-width intervals
    - Equal-frequency intervals
  - Converting categorical variables to numerical
    - Ordinal encoding
    - One-hot encoding
- **Feature Scaling -** *Why?*
  - Normalization/Min-Max scaling
  - Standardization
- Sampling strategies
  - Random/Stratified Sampling
  - Cross-validation
- Handling missing data
- Data processing pipelines

Chapter 3 - Model Evaluation

- Why is it necessary?
- **Binary Classification**
  - Confusion Matrix
  - Accuracy vs. Error Rate
    - Class Imbalance Problem
  - Precision vs. Recall vs. F1-score
  - Sensitivity vs. Specificity
  - Precision-Recall Curve
  - ROC / AUC
- **Multi-class Evaluation**
  - One-versus-All (OvA)
  - One-versus-One (OvO)
- Multi-label Classification
  - What's the difference between?

Chapter 4 - Linear Regression

- Simple vs. Multiple
- Regression Performance Measure
- **Gradient Descent**
  - Batch Gradient Descent
  - Stochastic Gradient Descent
  - Mini-Batch Gradient Descent
- Polynomial Regression
- Learning Curves
  - Bias/Variance Trade-Off
- Regularized Linear Models
  - Ridge vs. Lasso vs. Elastic Net
- Logistic Regression

Chapter 5 - Support Vector Machines (SVM)

- Linear SVM Classification
  - Margin/Support vectors
  - Hard vs. Soft Margin Classification
  - Dealing with Imbalanced Data
- Nonlinear SVM Classification
  - Kernel Trick - *What is it?*
  - Kernel Functions
    - Polynomial Kernel
    - Gaussian RBF Kernel
    - Similarity Features
- SVM Regression

Note: Detailed understanding of the math is NOT required :)